

A Design Space Exploration and Evaluation for Main-Memory Hash Joins in Storage Class Memory

Wentao Huang^{*}, Yunhong Ji[†], Xuan Zhou[‡], Bingsheng He^{*}, Kian-Lee Tan^{*}

^{*}National University of Singapore

[†]Renmin University of China

[‡]East China Normal University

Aug 31, 2023

01 The Memory Scaling Challenge

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 - Scaling DRAM technologies to sub-20nm is challenging

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 - Scaling DRAM technologies to sub-20nm is challenging
- Growing imbalance between memory demand and supply
 - Increasing memory demand in datacenter applications
 - DRAM capacity per core dropping by 30% biannually
- Approaches to break the wall
 - Manufacturing advancement for scaling DRAM
 - Exploring alternative technologies for DRAM

02

Storage Class Memories (SCMs)

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 - Good scaling ability
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 - Low cost per byte
 - DRAM-scale latency/bandwidth
 - Byte addressability
 - Data persistence

03

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- NVDIMM-P primitives



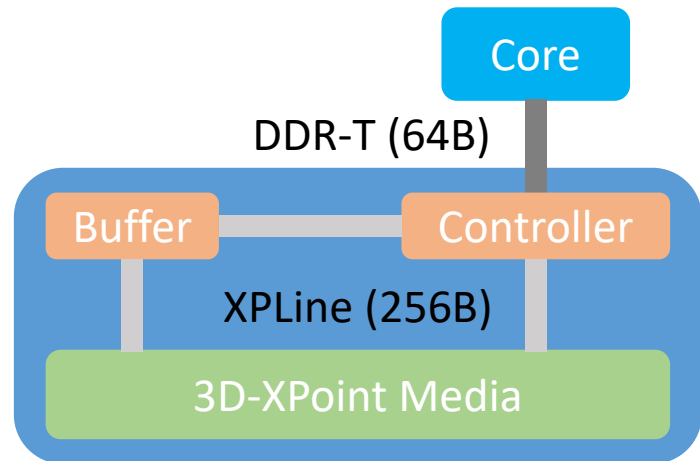
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- NVDIMM-P primitives
 - p_i : access granularity mismatch



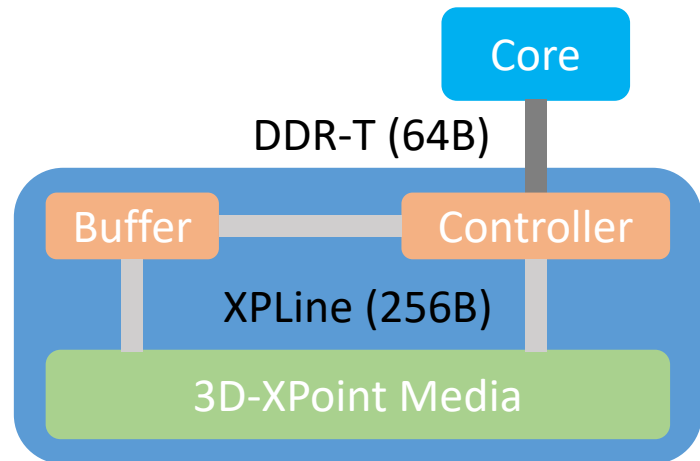
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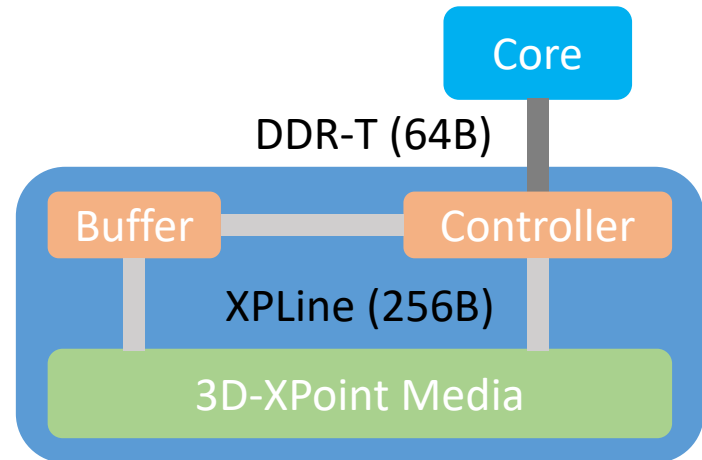
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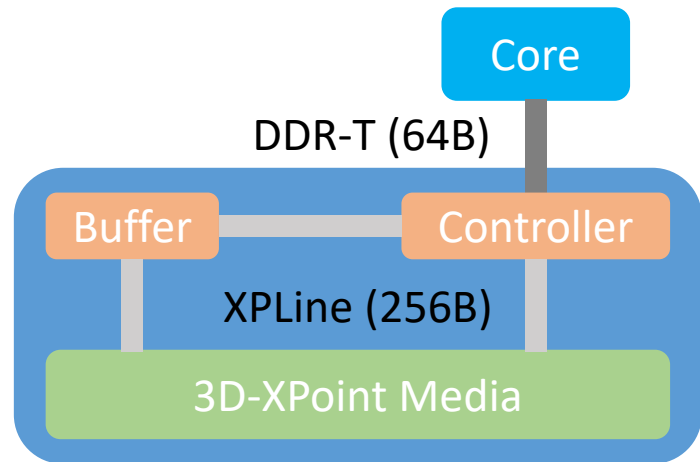
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- P_4 : costly page fault handling
- P_5 : persistent instructions supported



04 Main-Memory Hash Joins

Main-Memory Hash Joins

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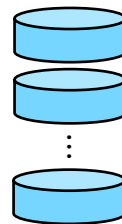
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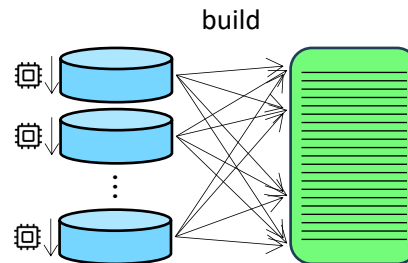
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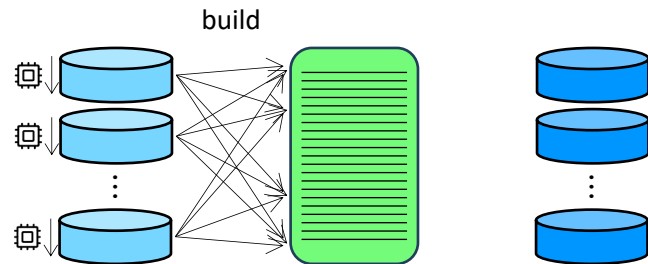
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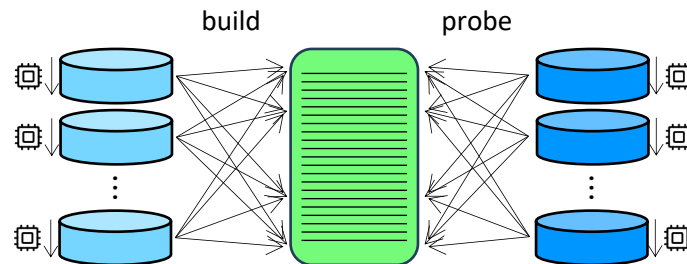
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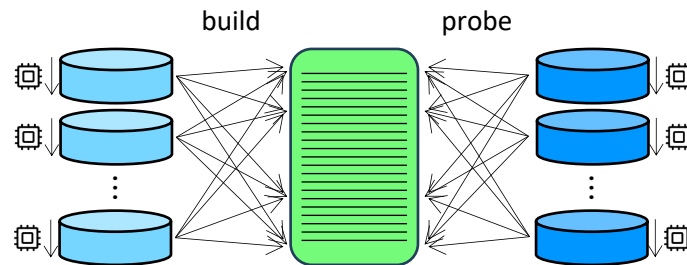
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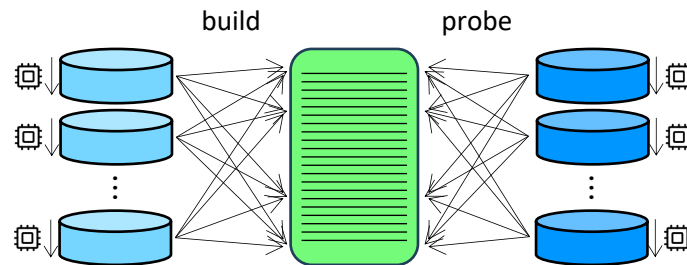
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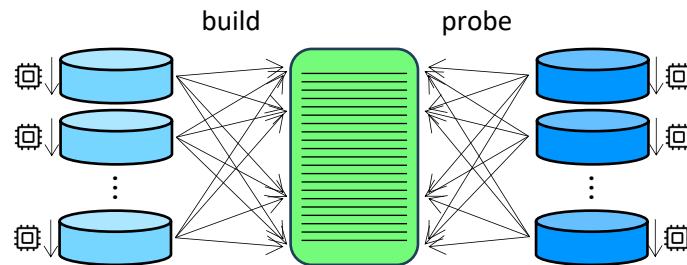
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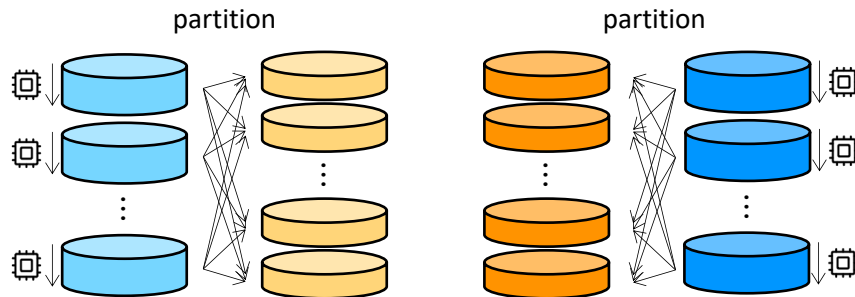
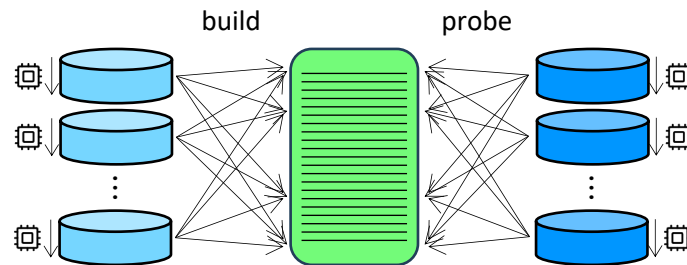
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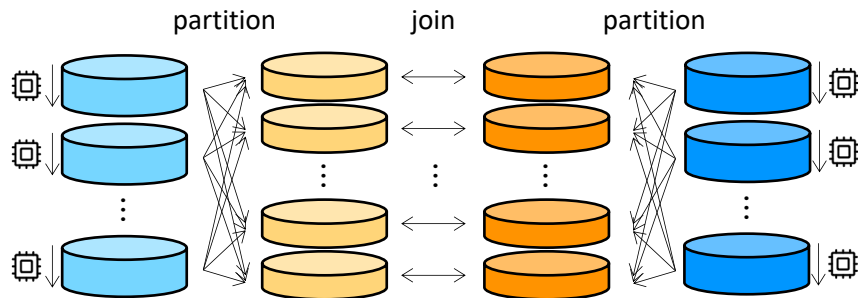
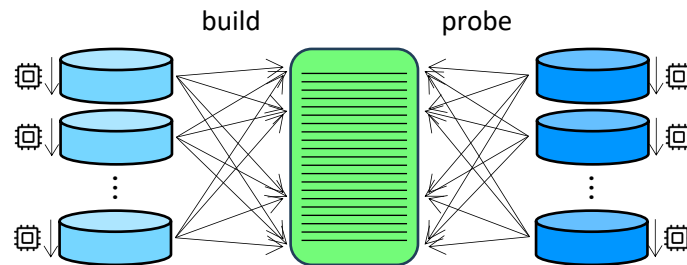
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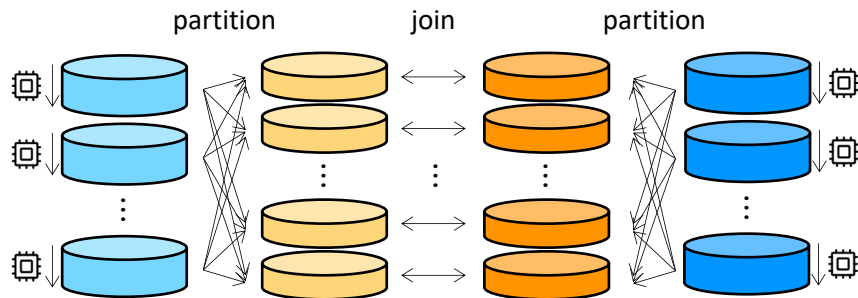
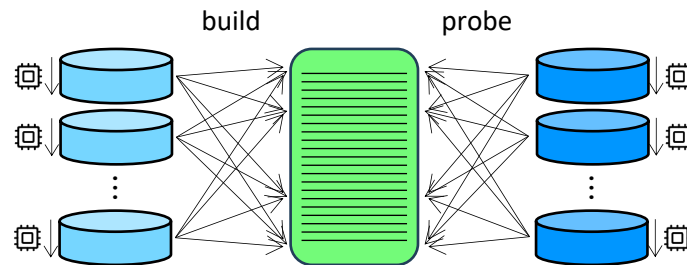
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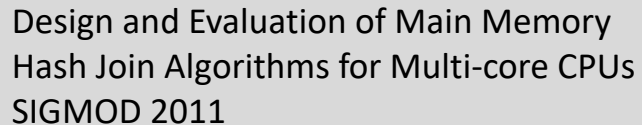
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 - Cache-sized scale join



05

An Ongoing Debate: PHJ vs. NPHJ

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An Ongoing Debate: PHJ vs. NPHJ



Design and Evaluation of Main Memory
Hash Join Algorithms for Multi-core CPUs



Main-Memory Hash Joins on Multi-Core
CPUs: Tuning to the Underlying Hardware
ICDE 2013

ETH zürich

▷ PHJ > NPHJ

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An Experimental Comparison of Thirteen
Relational Equi-Joins in Main Memory
SIGMOD 2016

▷ PHJ > NPHJ



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To Partition, or Not to Partition, That is the
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▷ NPHJ > PHJ



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Evaluating In-Memory Hash-Joins on Persistent Memory
EDBT 2022

▷ NPHJ > PHJ



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A Design Space Exploration and Evaluation for Main-Memory Hash Joins in Storage Class Memory VLDB 2023

▷ PHJ > NPHJ



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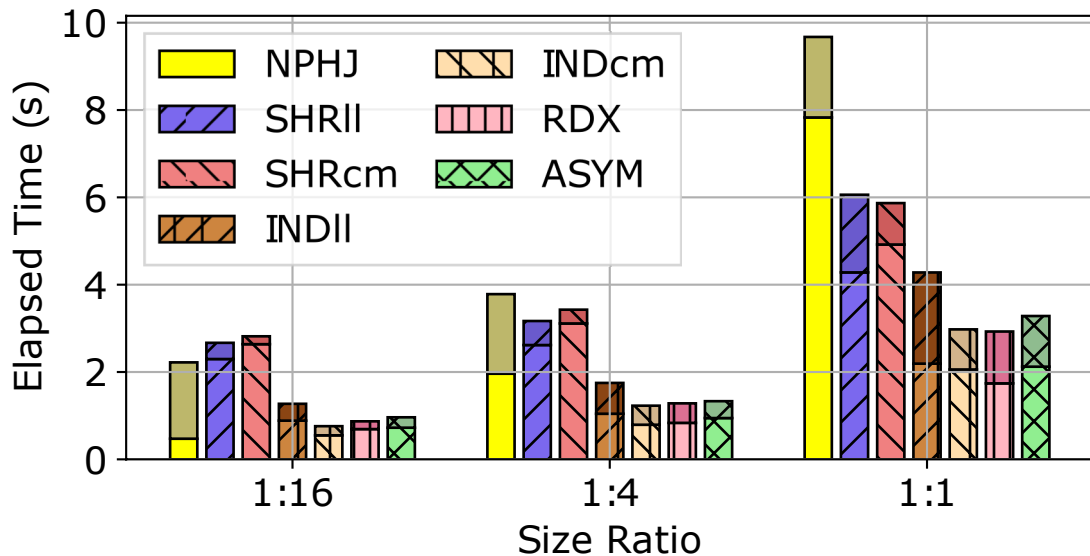
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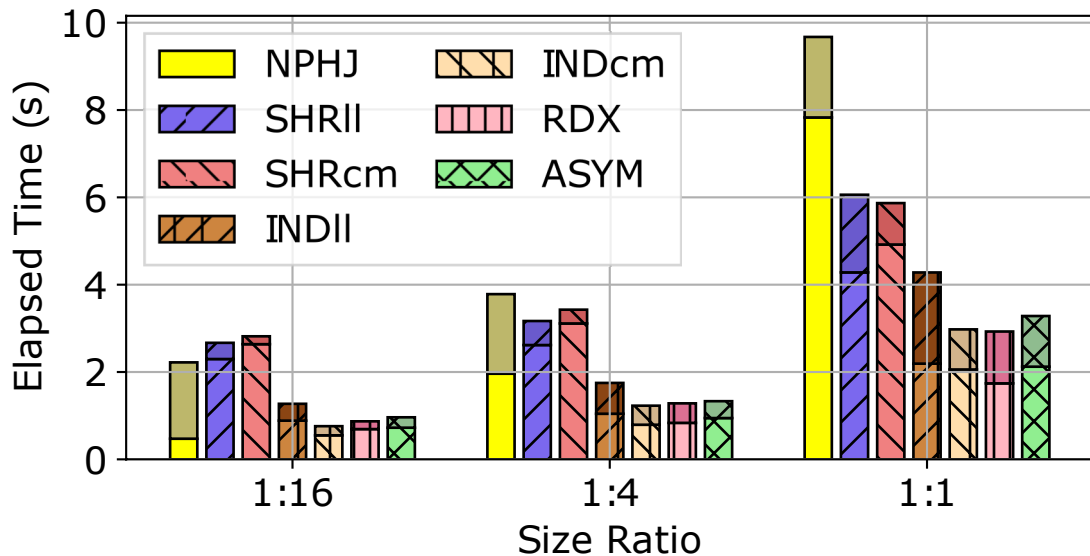
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PHJ vs. NPHJ: An Overall Comparison

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▷ PHJs generally outperform NPHJs

08

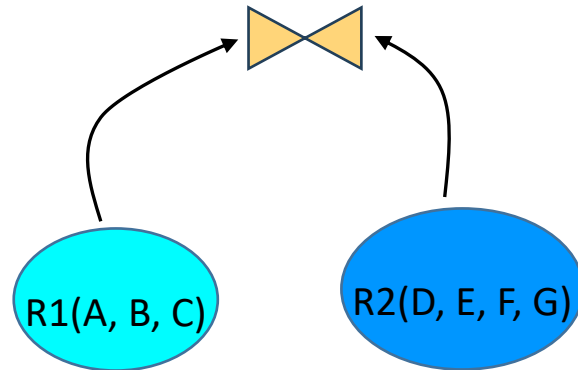
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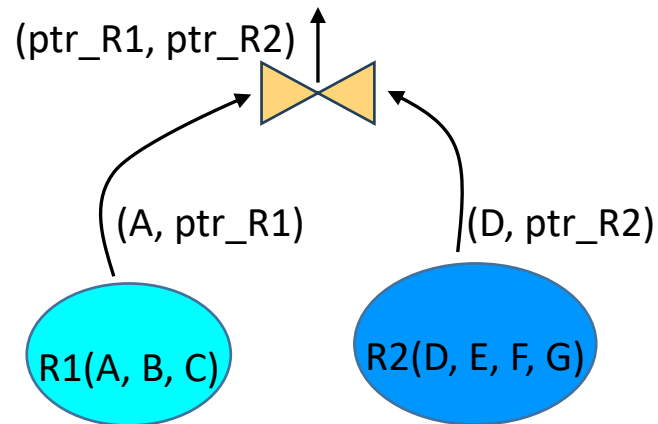
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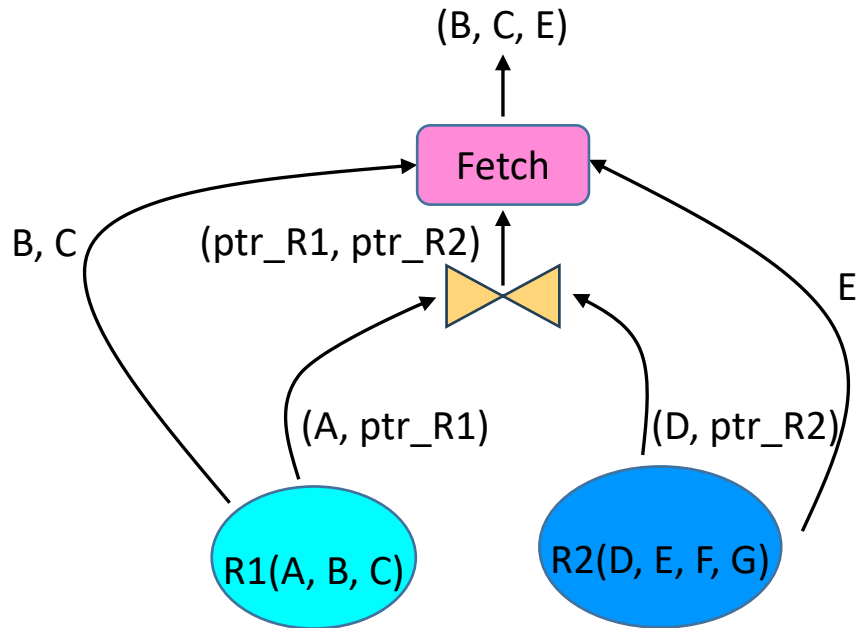
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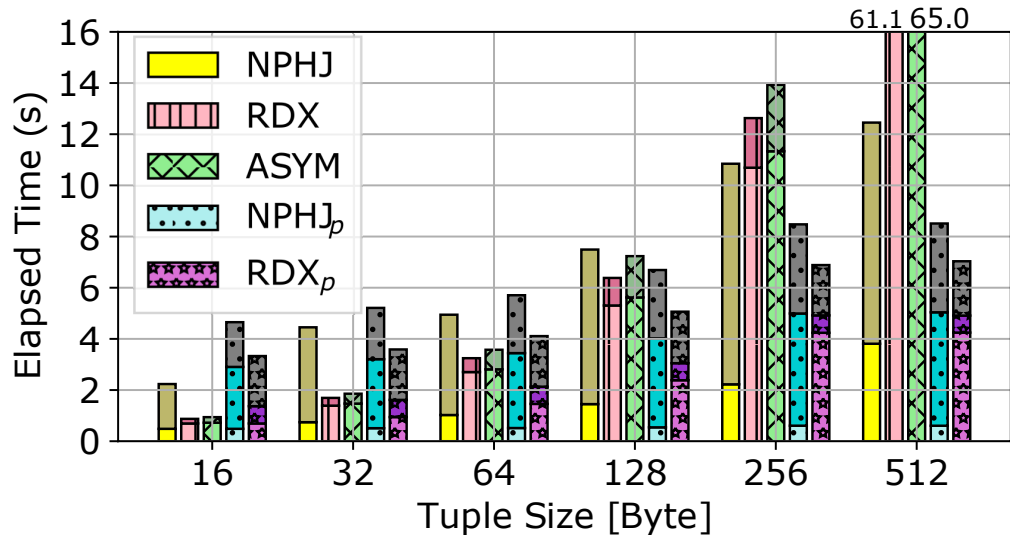
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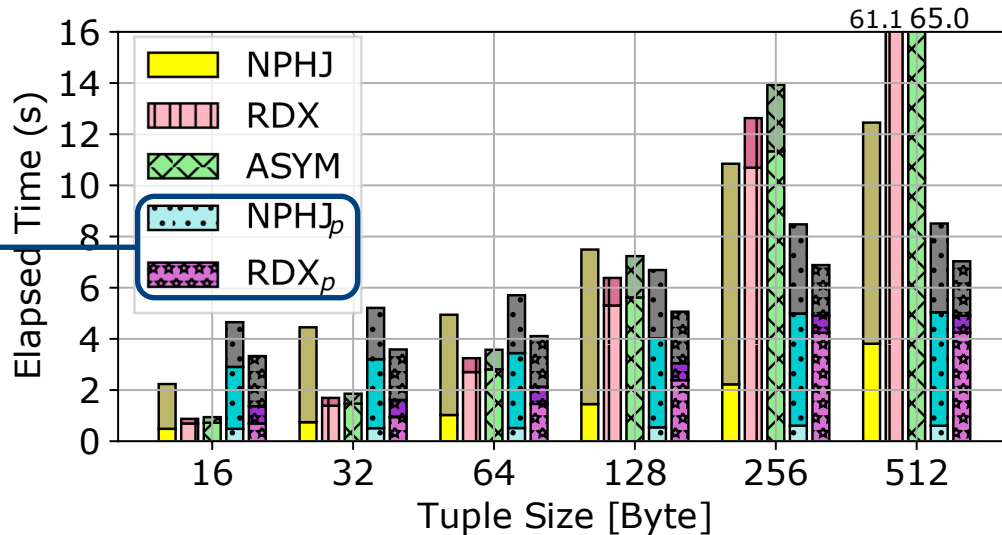
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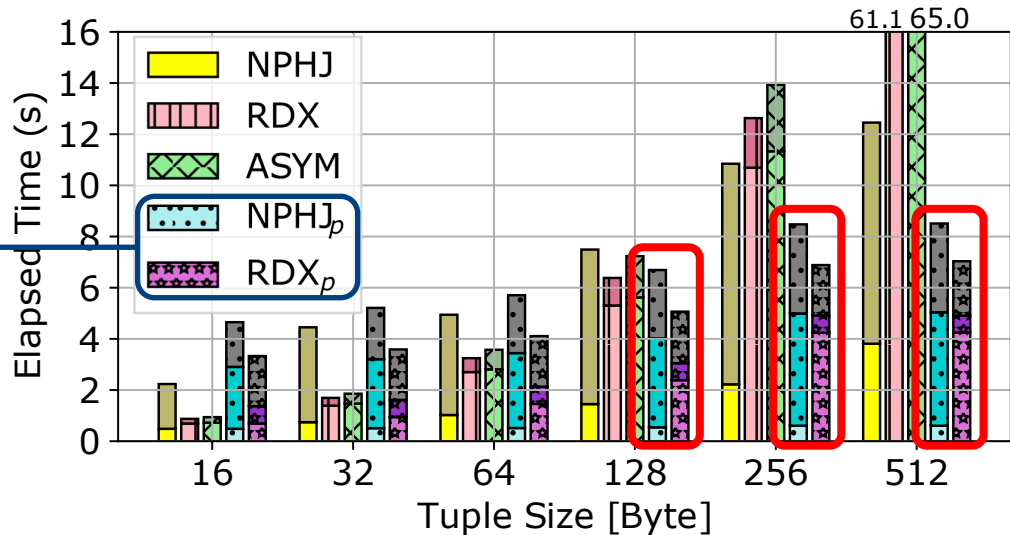
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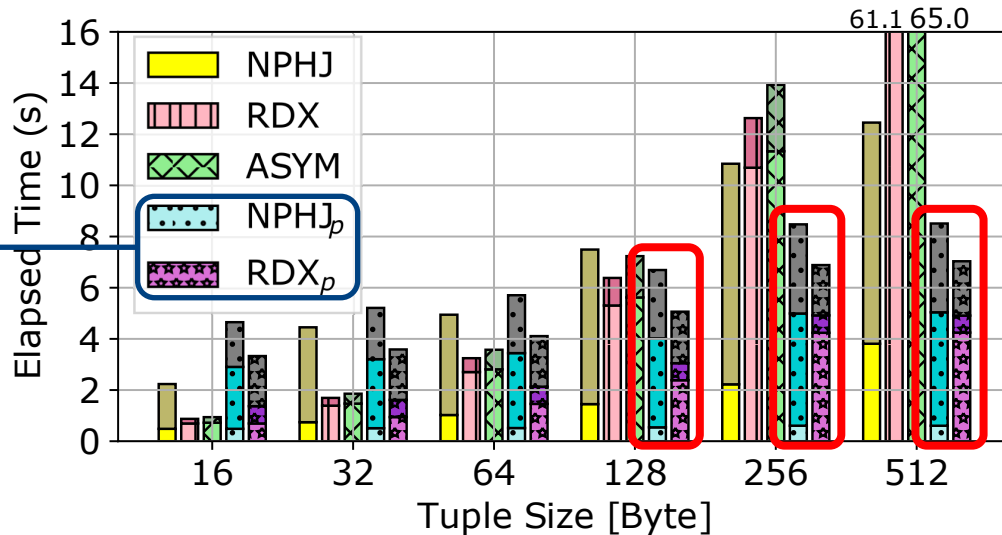


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<key, pointer>

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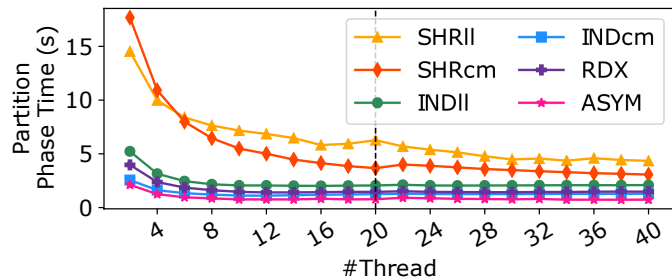


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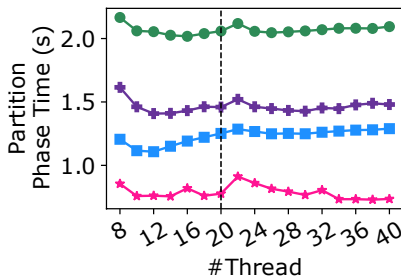
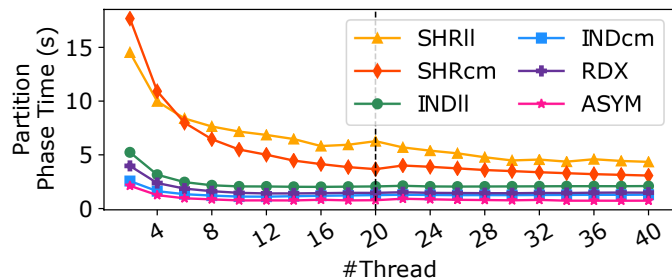
▷ Late materialization (*<key, pointer>*) -based join processing further reinforces PHJ's superiority against NPHJ

Bandwidth Regulation in Partitioning

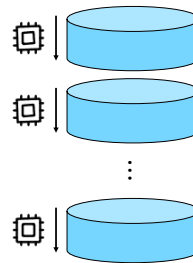
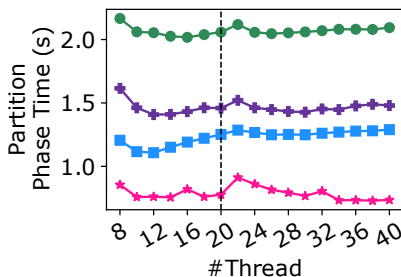
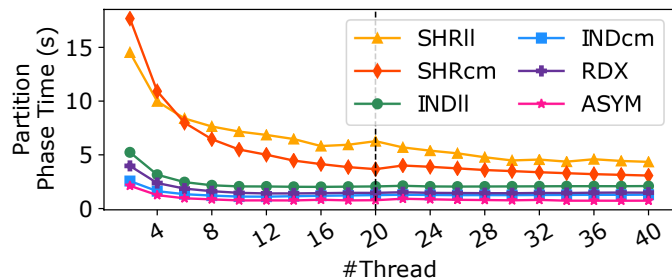
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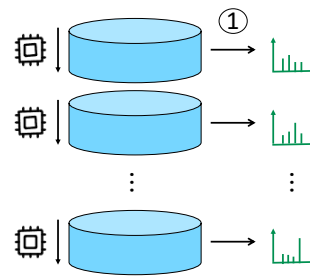
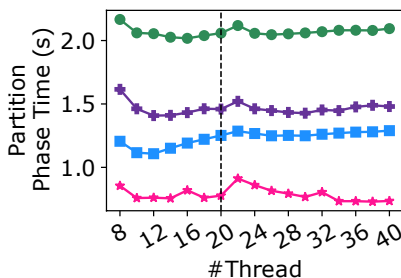
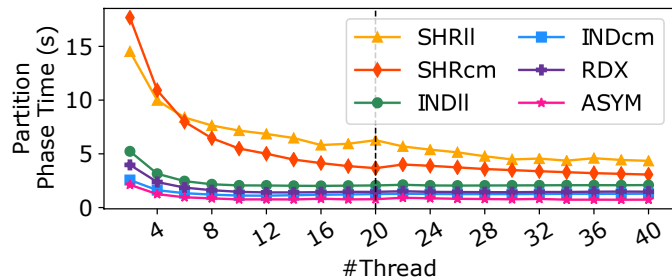
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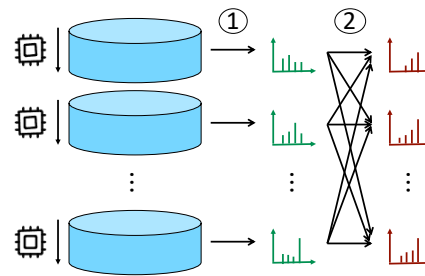
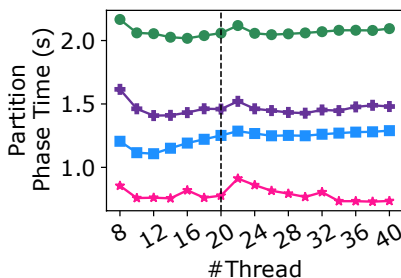
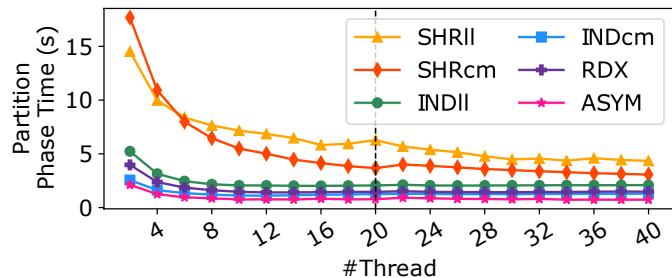
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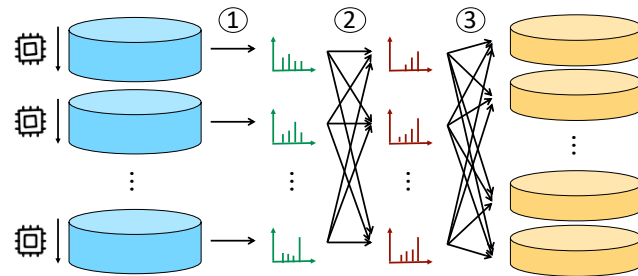
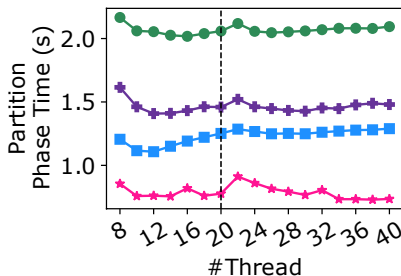
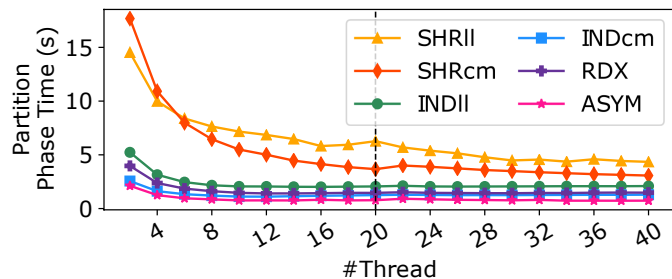
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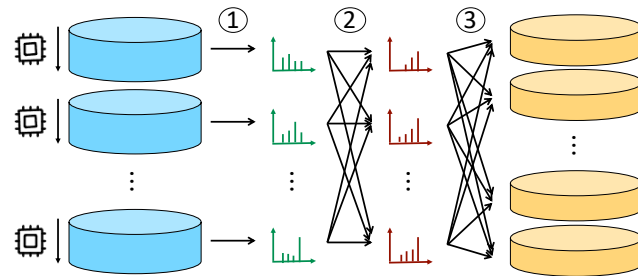
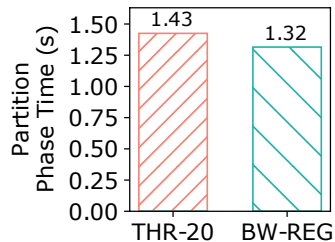
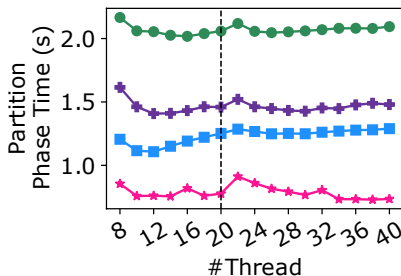
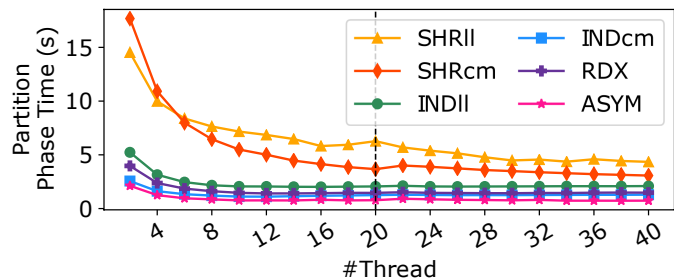
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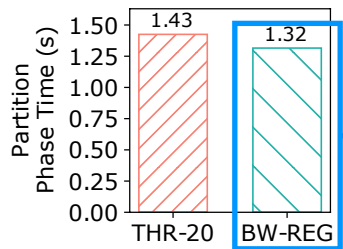
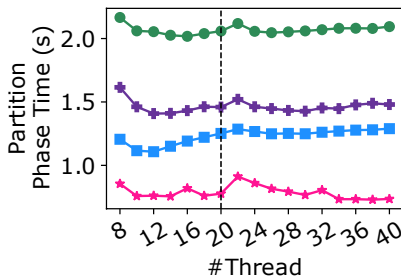
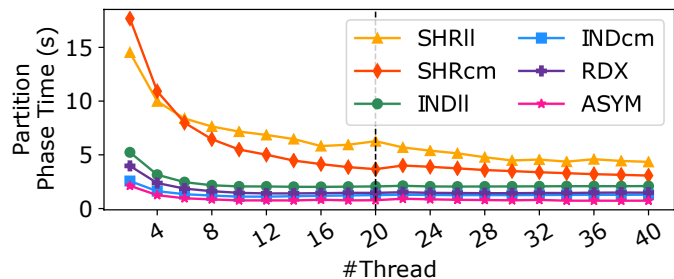
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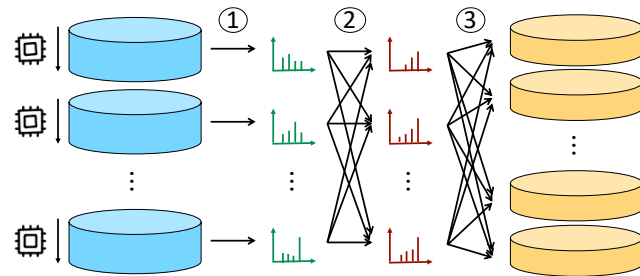
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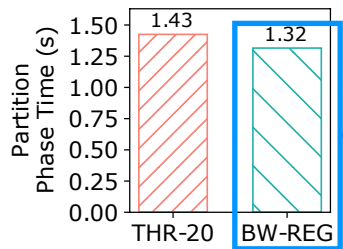
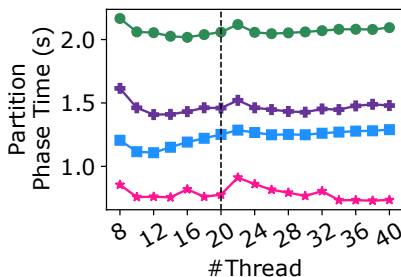
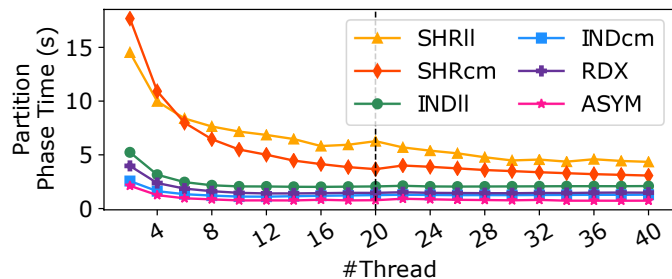
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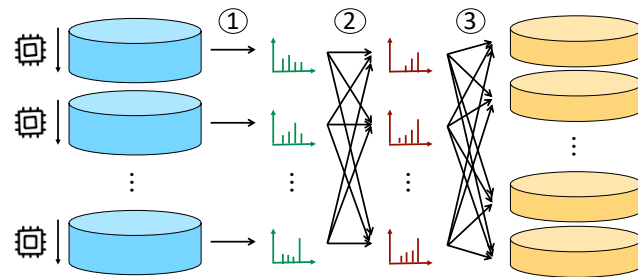
20-thread processing for step ① and ②
10-thread processing for step ③



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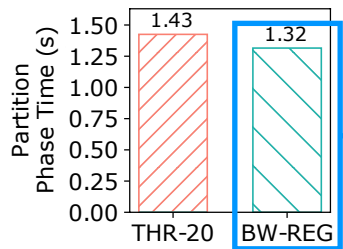
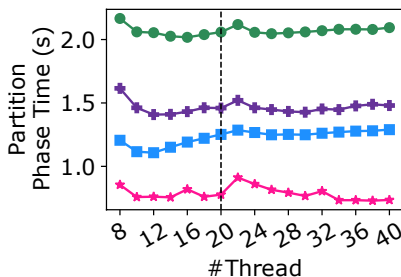
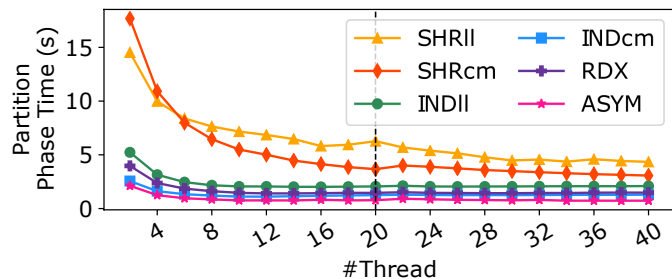


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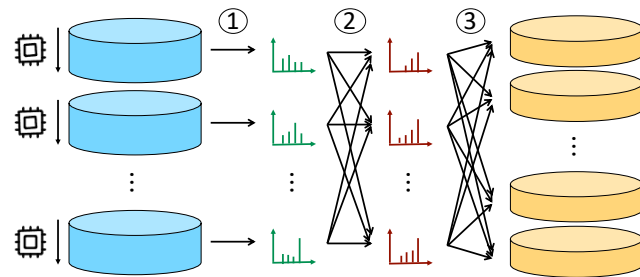


► Paralellism limitation in write-intensive steps further facilitates partitioning

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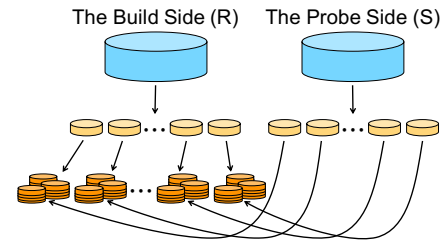
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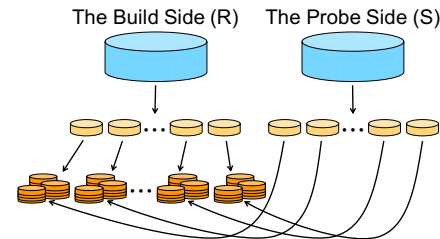
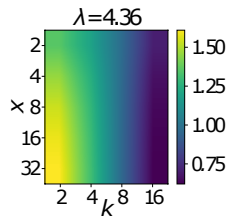
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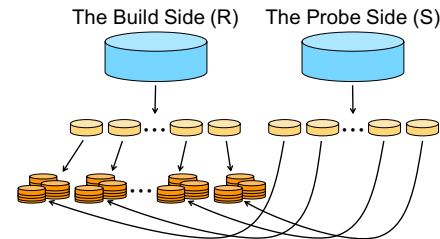
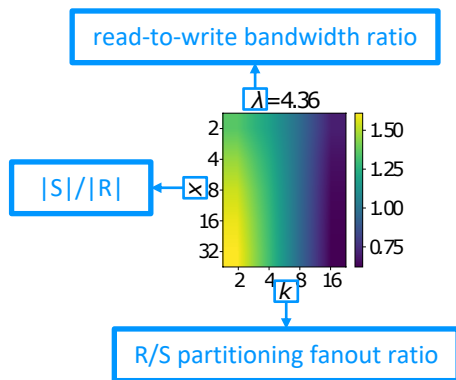
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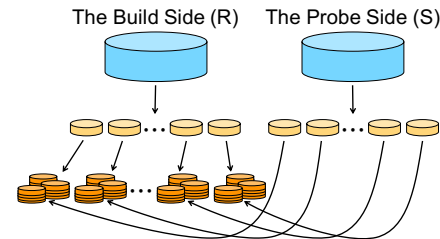
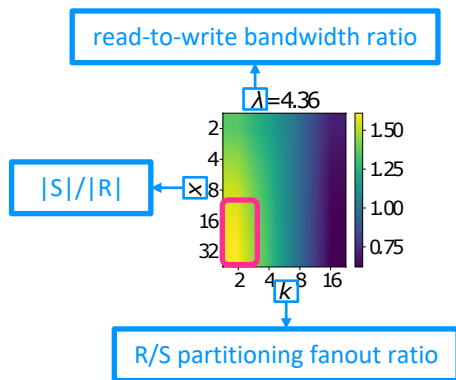
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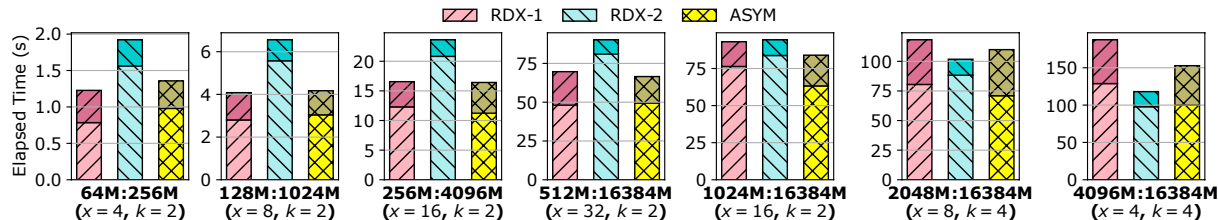
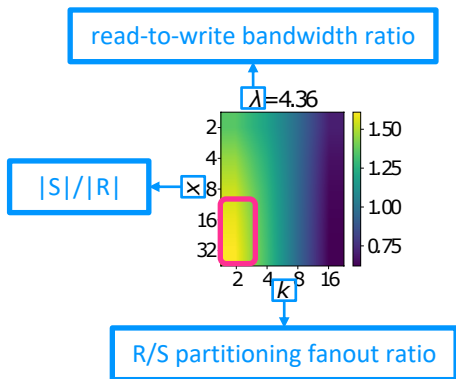
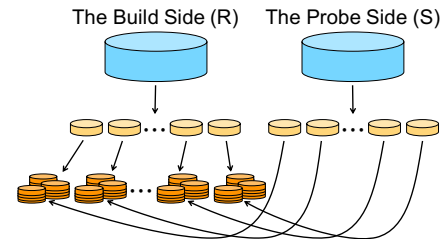
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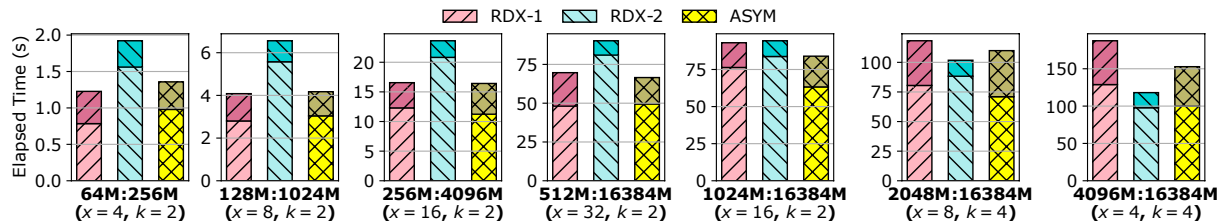
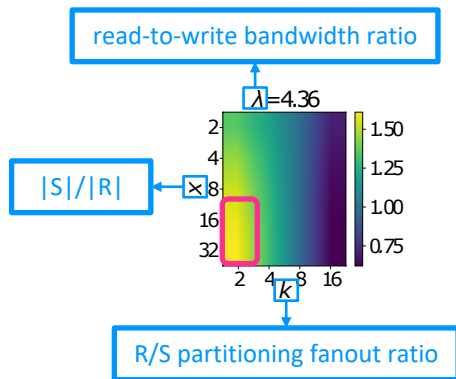
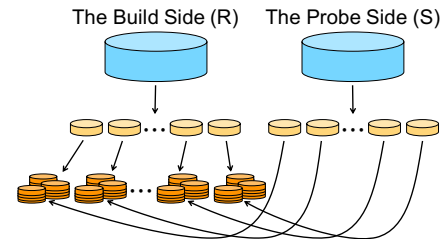
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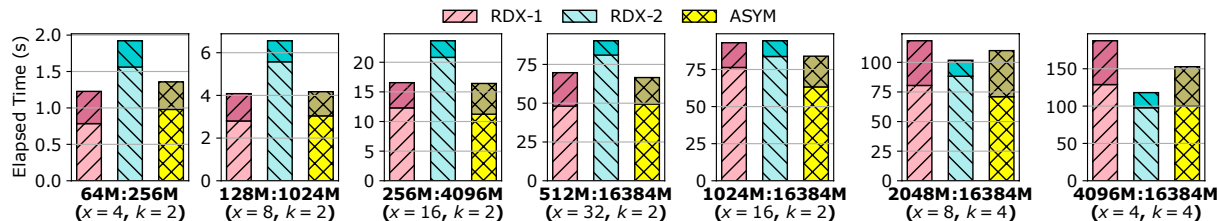
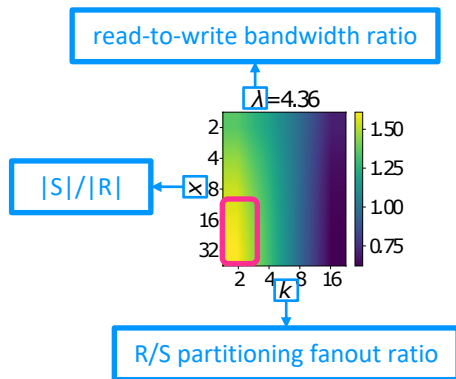
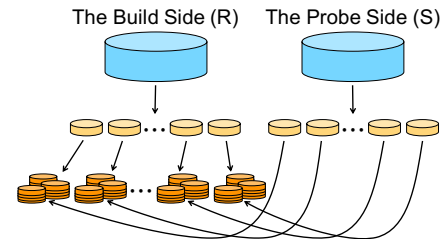
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- Additional experiments & analyses
 - Technical report: <https://nus.edu/3qIVvjJ>
 - Source codes: <https://github.com/fukien/hashjoin-scm>



THANK YOU